



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

DIGITAL FORENSICS

System Forensics, Investigation, and Response

Lesson 3

Forensics Methods and Labs

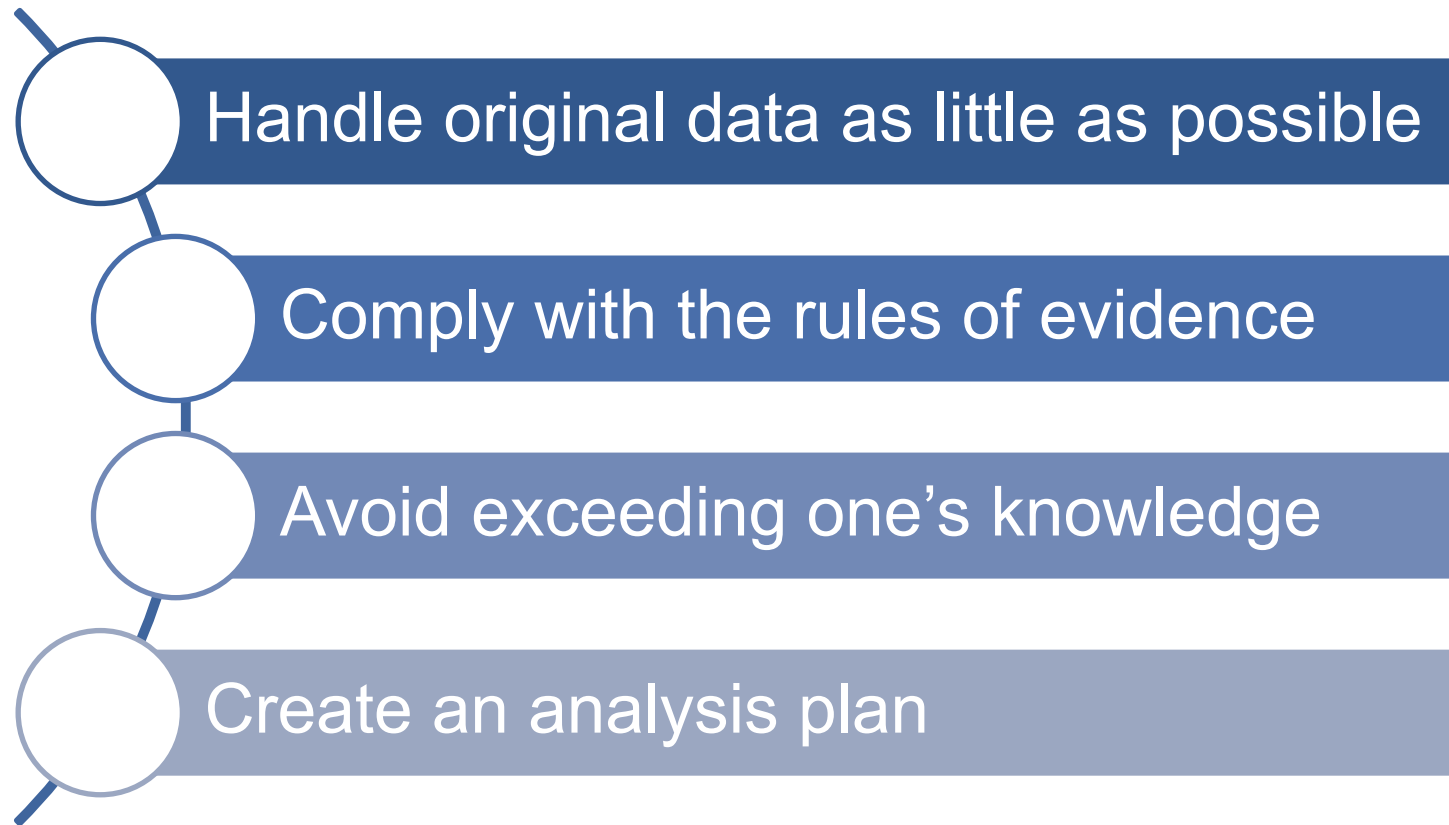
Learning Objective

- Describe digital forensic methodology and labs.

Key Concepts

- Forensic lab set-up
- Methodologies and approaches used in forensic investigations
- Evidence-handling tasks
- Common forensic software programs
- Documentation of methodologies and findings

Forensic Investigation Methodologies



Technical Information Collection Considerations

- Consider the life span of the information
 - Data is volatile
- Collect information quickly
- Collect bit-level information

Formal Forensic Approaches



Documentation

- Strong evidence-processing documentation
- Good chain-of-custody procedures
- A systems forensics specialist should have a good understanding of:
 - Computer hard disks and CDs, and know how to find hidden data in obscure places
 - The techniques and automated tools used to capture and evaluate file slack or slack space

File Slack Searching

- If you write a 1-kilobyte (KB) file to a disk that has a cluster size of 4 KB, the last 3 KB of the cluster are wasted
- This unused space between the logical end of file and the physical end of file is known as **file slack or slack space**
- File slack is a source of potential security leaks involving passwords, network logons, email, database entries, images, and word processing documents

Evidence-Handling Tasks

Find

- Gather evidence

Preserve

- Handle computers and storage media with caution

Prepare

- Document evidence source and ensure evidence has not changed

Evidence-Gathering Measures

Avoid changing evidence

Determine when evidence was created

Trust only physical evidence

Search throughout a device

Present evidence well

Expert Reports

- Formal documents that detail experts' findings
- Considerations include:

Format of the
report

Thoroughness

Back up
everything you
say

How to Set Up a Forensic Lab

Identify functions to be performed

Define activities and estimate workload

Determine necessary equipment and software

Determine physical space requirements

Plan for security

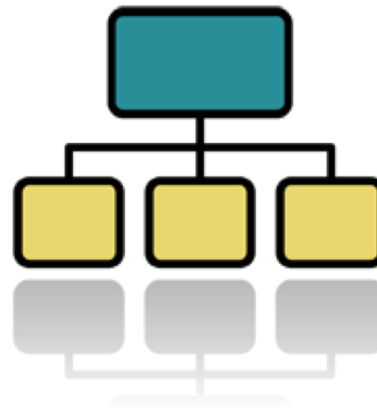
Equipment

- Computers
 - Server should have RAID 1 at a minimum
- Hard drives and storage
 - USB, SCSI, etc.
- Legacy and state-of-the-art



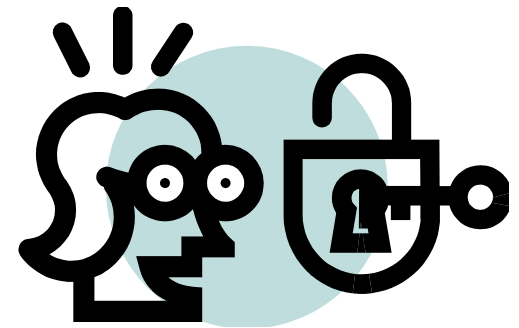
Equipment (Cont.)

- Peripherals
- Networking equipment
- Cables, adapters, and converters
- Write blockers
- Tools



Security

- Network and electronic security
- Lab network should not be attached to the Internet
- Includes physical security
 - Access to the lab
 - Ways of securing evidence



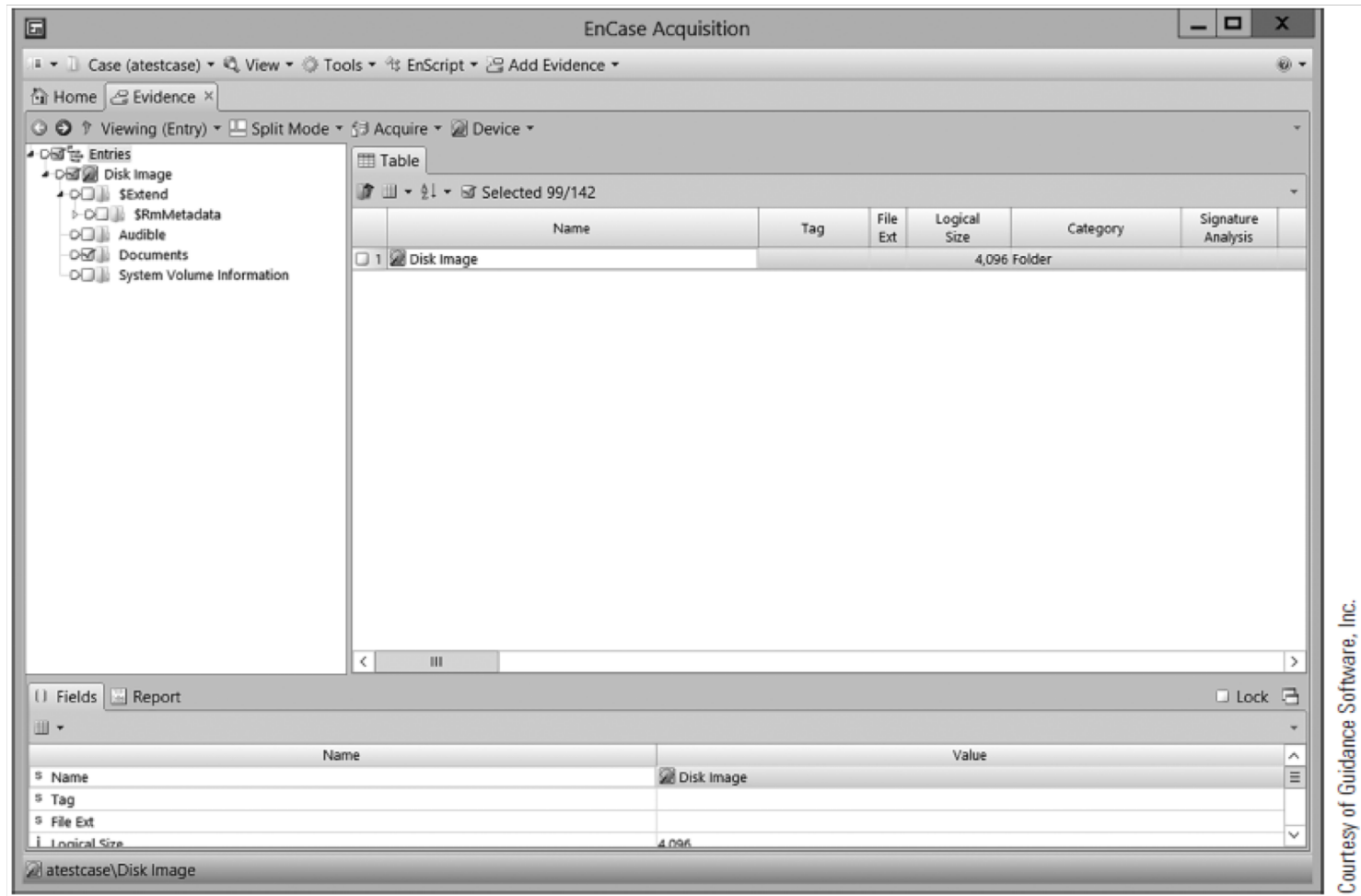
American Society of Crime Laboratory Directors (ASCLD)

- Provides guidelines for:
 - Managing a forensics lab
 - Acquiring crime lab and forensic lab certification
- A lab must meet about 400 criteria to achieve accreditation
- TEMPEST

Forensics Software Tools

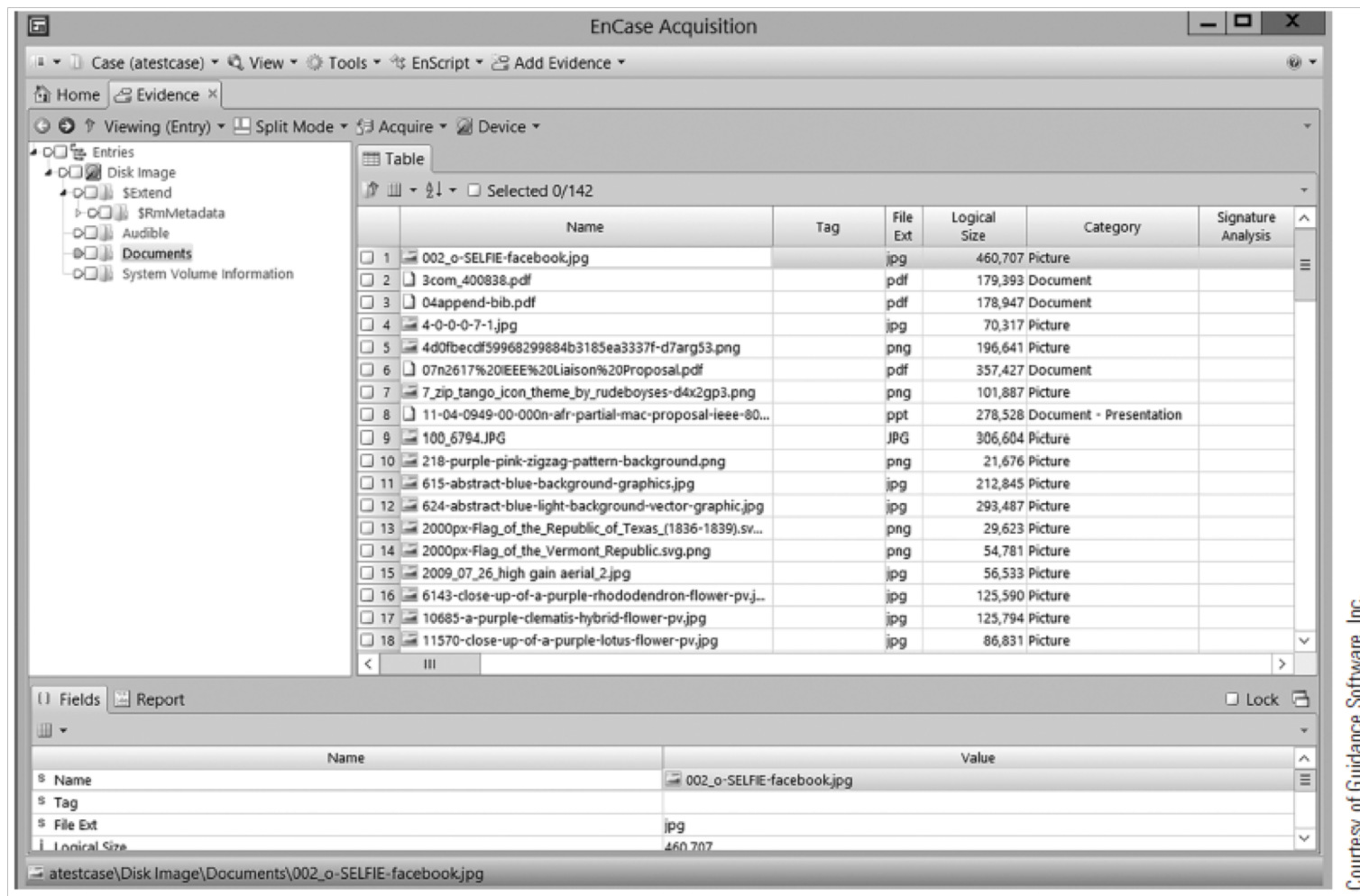
- EnCase
- Forensic Toolkit (FTK)
- OSForensics
- Helix
- Kali Linux
- AnaDisk disk analysis tool
- CopyQM Plus disk duplication software
- The Sleuth Kit
- Disk Investigator

EnCase Case File



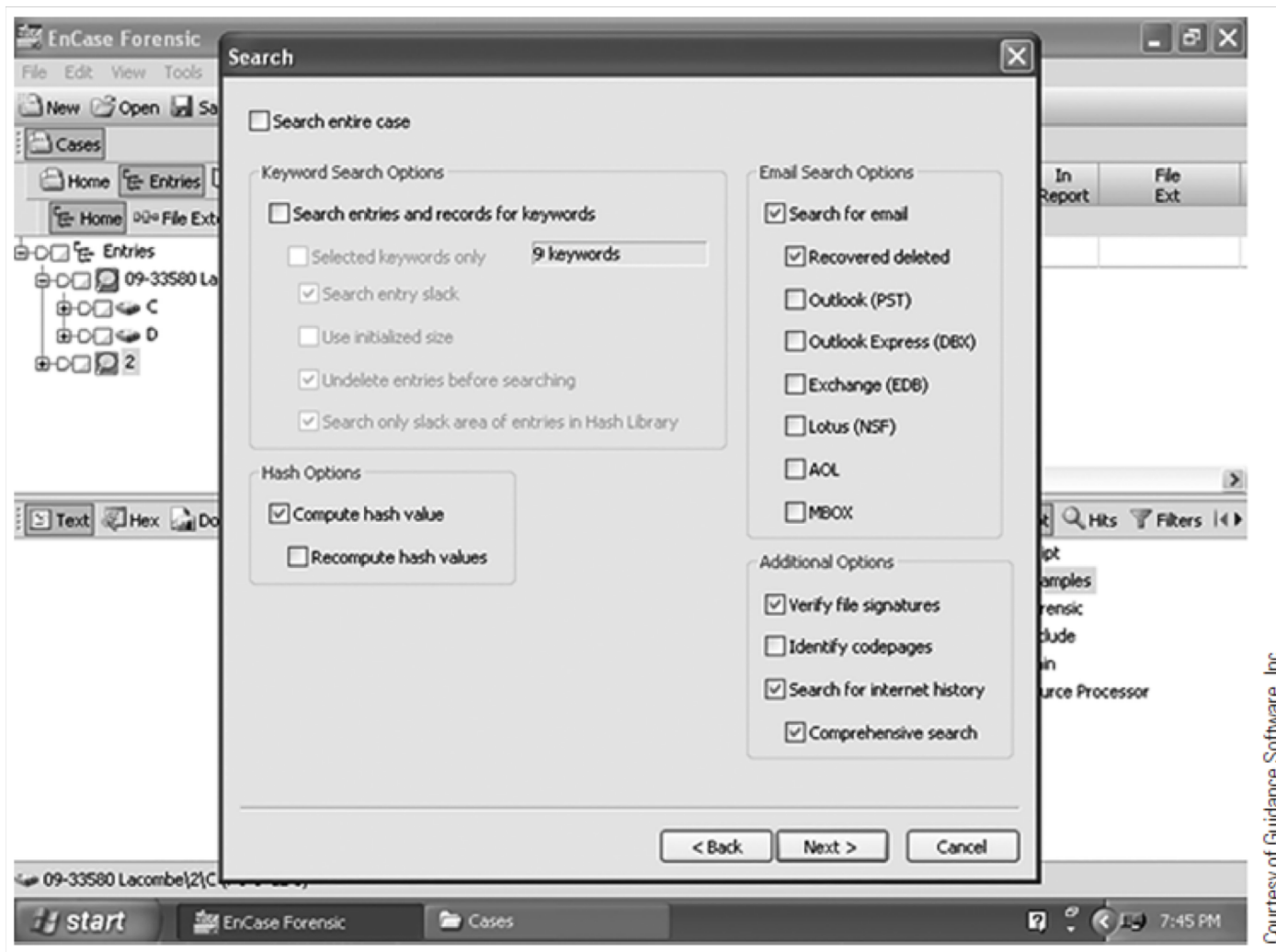
Courtesy of Guidance Software, Inc.

EnCase View Pane



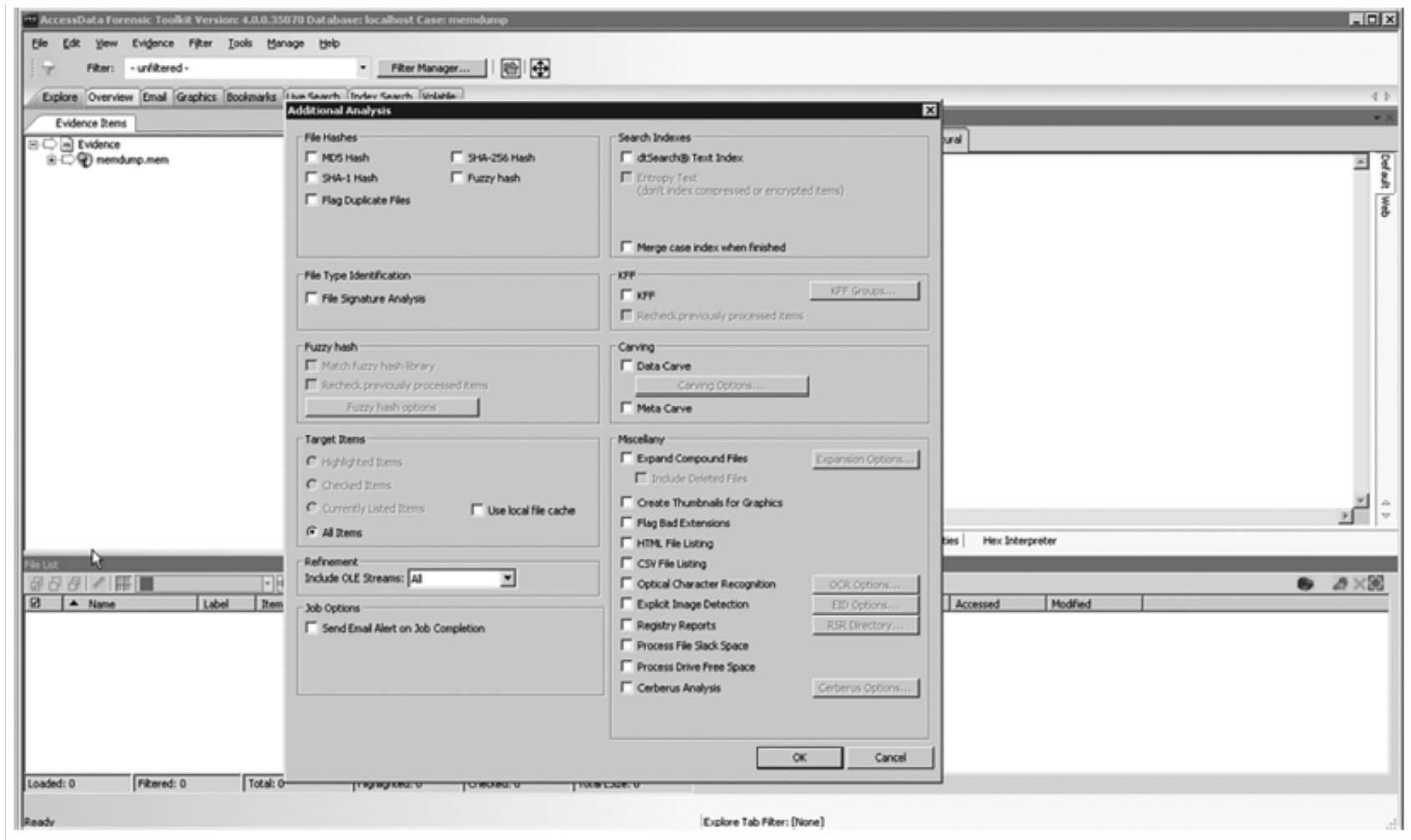
Courtesy of Guidance Software, Inc.

EnCase View Search



Courtesy of Guidance Software, Inc.

FTK Features



FTK Analysis



Courtesy of AccessData Group, Inc.

Certifications

PC hardware

- CompTIA A+

Basic networking

- CompTIA Network+ or the Cisco Certified Network Associate (CCNA)

Certifications

Security

- CompTIA Security+ and/or (ISC)2 CISSP certification

Hacking

- EC-Council Certified Ethical Hacker

Forensics-Specific Certifications

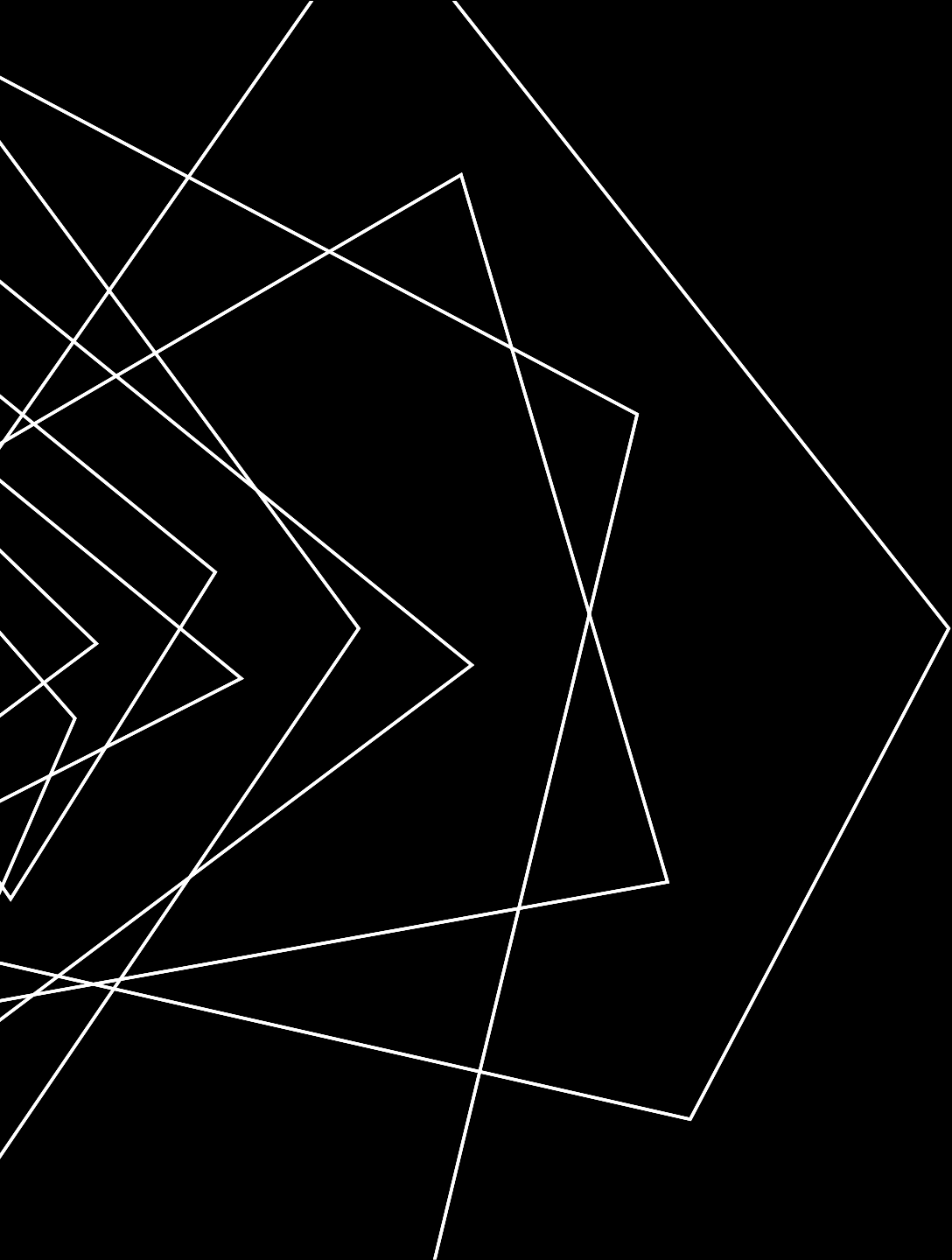


Forensics-Specific Certifications



Summary

- Forensic lab set-up
- Methodologies and approaches used in forensic investigations
- Evidence-handling tasks
- Common forensic software programs
- Documentation of methodologies and findings



THANK YOU

Yau Ka Cheung
Yauka.Cheung@etamu.edu
ACB1 Room 335